Data-Driven Framework for Credit Lending

1. The Market Opportunity

India's credit market presents a significant growth opportunity. The credit card industry has expanded at a

20% compound annual growth rate (CAGR) over the last five years, yet the market remains vastly underserved, with **only 3% of the population holding a formal credit card**. This gap is most pronounced in

Tier II and III cities and among **self-employed professionals**—segments that traditional lenders have found difficult to underwrite. Our fintech venture is strategically positioned to capture this opportunity by leveraging technology and a superior data-driven approach to serve these high-potential customers.

---

2. Our Data-Driven Solution for Loan Approval

To approve loans accurately and efficiently, we have developed a **Random Forest machine learning model** that predicts an applicant's probability of default with **81.8% accuracy**. The model's decisions are primarily driven by three key factors:

- **Credit History:** An applicant's past repayment behavior is the most critical predictor.
- **Total Income:** The combined income of the applicant and any co-applicant.
- **Loan Amount:** The size of the requested loan relative to income.

This data-driven underwriting process, grounded in the principles of the

5Cs of Credit, allows us to automate the majority of our approval decisions, ensuring a fast, consistent, and reliable experience for our customers.

---

3. Our Robust Risk Management Framework

Beyond individual loan approvals, we have established a comprehensive framework to manage portfolio-wide risk. Our analysis of a hypothetical $100 Million loan portfolio yielded two key insights:

- Under normal market conditions, our one-year
- **Value at Risk (VaR)** is **$8.2 Million** at a 95% confidence level, providing a clear baseline for our capital reserves.
- A
- **macroeconomic stress test**, simulating a severe recession modeled after the 2008 financial crisis, revealed that our portfolio's expected losses could surge to **$11.5 Million**. The test identified a significant vulnerability to downturns in the
- **Real Estate & Construction sector**.

Based on this, we will implement strict risk controls, including **sector concentration limits** and **additional capital buffers**, to ensure our portfolio remains resilient even during severe economic shocks.

---

4. Our Customer-Centric Strategy

Our strategy extends beyond analytics and risk to focus on building long-term customer loyalty. An analysis of customer feedback revealed that nearly

50% of support tickets are driven by confusion around billing terms, particularly **minimum payments** and how interest accrues on revolving credit. To address this, we will launch a

proactive communication initiative, using personalized in-app notifications to educate customers on their payment options. This approach is designed to reduce customer churn, improve financial literacy, and build trust.

---

5. Final Strategic Roadmap

To execute this vision, we recommend the following strategic initiatives for the next 12 months:

- **Deploy the Random Forest Model:** Immediately begin using the model to automate loan approvals, reducing decision times and improving operational efficiency.
- **Launch Targeted Acquisition Campaigns:** Focus marketing and business development efforts on the identified high-potential semi-urban customer segment.
- **Implement Portfolio Risk Controls:** Formally establish the recommended sector concentration limits and increase capital reserves to align with stress test findings.
- **Roll Out a Customer Education Initiative:** Develop and launch the proactive communications program to reduce billing-related queries and enhance customer satisfaction.